

上线指令(02H)

设备发送(02H)

设备通过jumi/deviceconn主题发布

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "02",
  "Ack": "00",
  "data": {
    "DevType": "xx",      //设备类型
    "Ver": "xxx",        //固件版本号
    "ContainNum": xx     //货柜数
    "LayNum": "xx",      //层架数 (行)
    "MotoNum": "xx",     //货道数 (列)
  }
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```

后台响应 (02H)

后台通过jumi/00000000000000000000000000000000主题发布(主题数字部分为机器编号)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "02",
  "Ack": "00",
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```

设备状态推送(08H)

定时10min推送

设备发送 (08H)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "08",
  "Ack": "00",
  "data": {
    "List": [
      {
        "contain": 0, // 货柜号
        "rackwidth": 6, // 列宽度
        "rackheight": 6, // 行高度
        "current": [
          [1, 1, 1, 1, 1, 1],
          [1, 1, 1, 1, 1, 1],
          [1, 1, 1, 1, 1, 1],
          [1, 1, 1, 1, 1, 1],
          [1, 1, 1, 1, 1, 1],
          [0, 0, 0, 0, 0, 0]
        ],
        "temp": 25, // 温度值
        "IrErr": 102000, // 错误码
      },
      ... // 其他货柜
    ]
    "Rssi": 135, // 信号强度值
    "NET": "LTE", // 信号网络
    "PowerStatus": xx, // 强电电源状态 1 供电 0 断电
    "BatteryLevel": xx, // 电池电量
    "longitude": "xx", // 经度
    "latitude": "xx" // 纬度
  }
  "Time": {
    "year": xx, // 年
    "month": xx, // 月
    "day": xx, // 日
    "hour": xx, // 时
    "minute": xx, // 分
    "second": xx, // 秒
  }
}
```

后台响应 (08H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "08",
  "Ack": "00",
  "Time": {
    "year": xx,      //年
    "month": xx,     //月
    "day": xx,       //日
    "hour": xx,      //时
    "minute": xx,    //分
    "second": xx,    //秒
  }
}
```

读取设备状态(03H)

后台下发(03H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "03",
  "Ack": "00",
  "data": {
    "ContainNum": xx, //货柜号
  }
  "Time": {
    "year": xx,      //年
    "month": xx,     //月
    "day": xx,       //日
    "hour": xx,      //时
    "minute": xx,    //分
    "second": xx,    //秒
  }
}
```

设备响应(03H)

设备通过jumi/devicecomm000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
```

```

"SN": "xxxxxxxxxxxxxxxx",
"Cmd": "03",
"Ack": "00",
"data": {
    "List": [
        {
            "contain": 0, // 货柜号
            "rackwidth": 6, // 列宽度
            "rackheight": 6, // 行高度
            "current": [
                [1,1,1,1,1,1],
                [1,1,1,1,1,1],
                [1,1,1,1,1,1],
                [1,1,1,1,1,1],
                [1,1,1,1,1,1],
                [0,0,0,0,0,0]
            ]
            "temp": xx, // 温度值
            "IrErr": 102000, // 错误码
        }
    ]
    "Rssi": "135", // 信号强度值
    "NET": "LTE", // 信号网络
    "PowerStatus": xx, // 强电电源状态 1 供电 0 断电
    "BatteryLevel": xx // 电池电量
    "longitude": "xx", // 经度
    "latitude": "xx" // 纬度
}
"Time": {
    "year": xx, // 年
    "month": xx, // 月
    "day": xx, // 日
    "hour": xx, // 时
    "minute": xx, // 分
    "second": xx, // 秒
}
}

```

出货指令(01H)

后台下发(01H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```

{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "01",
  "Ack": "01",
  "data": {
    "List": [
      {

```

```

        "ContainNum":x,      //货柜号
        "MotoNum":x,        //货道号
        "LayNum":x,         //层架号
    },
    {
        "ContainNum":x,      //货柜号
        "MotoNum":x,        //货道号
        "LayNum":x,         //层架号
    },
    ...
]
}
"Time":{
    "year":xx,              //年
    "month":xx,             //月
    "day":xx,               //日
    "hour":xx,              //时
    "minute":xx,            //分
    "second":xx,            //秒
}
}

```

设备响应(01H,ACK:00)

设备通过jumi/servercomm0000000000000001主题发布(该主题通过20H指令获得)

```

{
  "ID":"xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN":"xxxxxxxxxxxxxxxx",
  "Cmd":"01",
  "Ack":"00",              //响应后台
  "data":
  {
    "AckError":xxxxxx      //113000 正常
                           //113301 消息过期
                           //113302 时间未同步
                           //113303 出货忙
                           //113304 内存分配失败
  }
  "Time":{
    "year":xx,              //年
    "month":xx,             //月
    "day":xx,               //日
    "hour":xx,              //时
    "minute":xx,            //分
    "second":xx,            //秒
  }
}

```

设备反馈出货结果(01H,ACK:01)

设备通过jumi/servercomm0000000000000001主题发布(该主题通过20H指令获得)

```

{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "01",
  "Ack": "01",          //设备主动发送结果
  "data": {
    "ContainNum": 0,    //货柜号
    "MotoNum": 1,       //货道号
    "LayNum": 1,        //层架号
    "Status": "xx",     //出货结果
    "MotorErr": 101000, //电机错误码
    "IrErr": 102000,    //红外错误码
  }
  "Time": {
    "year": xx,         //年
    "month": xx,        //月
    "day": xx,          //日
    "hour": xx,         //时
    "minute": xx,       //分
    "second": xx,       //秒
  }
}

```

设置设备温控参数(04H)

后台下发(04H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为)

```

{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "04",
  "Ack": "00",
  "data": {
    "ContainNum": 0,    //货柜号
    "En": "xx",        //使能
    "Mod": "xx",       //模式
    "Temp": xx,         //温度
  }
  "Time": {
    "year": xx,         //年
    "month": xx,        //月
    "day": xx,          //日
    "hour": xx,         //时
    "minute": xx,       //分
    "second": xx,       //秒
  }
}

```

设备响应

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xx xx xx xx xx xx xx xx",
  "Cmd": "04",
  "SN": "xx xx xx xx xx xx xx xx",
  "Ack": "00",
  "data": {
    "ContainNum": 0, // 货柜号
  }
  "Time": {
    "year": xx, // 年
    "month": xx, // 月
    "day": xx, // 日
    "hour": xx, // 时
    "minute": xx, // 分
    "second": xx, // 秒
  }
}
```

电子锁开门(09H)

后台下发(09H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "09",
  "Ack": "00",
  "data": {
    "ContainNum": xx, // 货柜号
    "En": "xx", // 使能
  }
  "Time": {
    "year": xx, // 年
    "month": xx, // 月
    "day": xx, // 日
    "hour": xx, // 时
    "minute": xx, // 分
    "second": xx, // 秒
  }
}
```

设备响应(09H,ACK00)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "09",
  "Ack": "00",           //设备响应
  "data": {
    "ContainNum": xx,    //货柜号
    "Status": "xx",     //01成功 02失败
  }
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```

设备反馈开锁结果(09H,ACK01)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",
  "Cmd": "09",
  "Ack": "01",           //返回结果
  "data": {
    "ContainNum": xx,    //货柜号
    "Status": "xx",     //开锁结果
  }
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```

设备首次连接(20H)

获取 业务主题 及 设备管理主题

设备向后台请求(20H)


```

"Cmd": "21",
"Ack": "00",
"Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
}
}

```

升降平台定位 (109H)

后台发送(109H)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```

{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",          //定长为16字节的字符串
  "Cmd": "109",
  "Ack": "00",
  "data": {
    "ContainNum": xx,          //货柜号
    "LayNum": xx,             //层架号 (层架号为 255 (即0xFF)时为设置出货口)
  }
  "Time": {
    "year": xx,              //年
    "month": xx,             //月
    "day": xx,               //日
    "hour": xx,              //时
    "minute": xx,            //分
    "second": xx,            //秒
  }
}

```

设备响应(109H)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```

{
  "ID": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "SN": "xxxxxxxxxxxxxxxx",          //定长为16字节的字符串
  "Cmd": "109",
  "Ack": "00",
  "data": {
    "Status": "xx",          //01成功 02失败
    "ContainNum": xx,        //货柜号
    "LayNum": xx,            //层架号 (层架号为 255 (即0xFF)时为设置出货口)
  }
}

```

```

    }
    "Time":{
        "year":xx,           //年
        "month":xx,         //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}

```

发起远程升级指令(0AH)

后台发送(0AH)

后台通过jumi/146xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx主题发布(该主题数字部分为机器编号)

```

{
    "ID": "xxxxxxxxxxxxxxxxxxxx"
    "Cmd": "0A"
    "Ack": "00"
    "SN": "xxxxxxxxxxxxxxxxxxxx"
    "data": {
        "PubTopic": "xx", //发布主题(例: jumi/deviceupgrade)
        "SubTopic": "xx", //订阅主题(例: jumi/upgrade)
        "Host": "xxxxx", //请求域名
        "URL": "xxxxx", //请求文件路径
        "FwVer": "Vx.x.x", //固件版本
        "FwSize": xx, //固件大小
        "FwMD5": "xxxxx" //固件MD5
    }
    "Time": {
        "year": xx, //年
        "month": xx, //月
        "day": xx, //日
        "hour": xx, //时
        "minute": xx, //分
        "second": xx, //秒
    }
}

```

设备响应(0AH,ACK00)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```

{
    "ID": "xxxxxxxxxxxxxxxxxxxx"
    "Cmd": "0A"
    "Ack": "00"
    "SN": "xxxxxxxxxxxxxxxxxxxx"
}

```

```

    "Time":{
        "year":xx,          //年
        "month":xx,        //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}

```

升级进度反馈(0AH,ACK01)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```

{
    "ID":"xxxxxxxxxxxxxxxxxx"
    "Cmd":"0A"
    "Ack":"01"
    "SN":"xxxxxxxxxxxxxxxxxx"
    "data":{
        "Ratio":xx,
            //升级进度
        "Status":xx,
            //0 升级开始
            //1 升级中
            //2 升级结束
        "Err":xx ,
            //120000 正常
            //120101 文件非法
            //120102 固件版本相同
            //120103 内存空间不足
            //120104 获取固件超时
            //120105 数据保存失败
            //120406 数据异常
    }
    "Time":{
        "year":xx,          //年
        "month":xx,        //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}

```

双向心跳(0BH & 1BH)

后台查询设备方向(0BH)

后台通过jumi/146xx主题发布(该主题数字部分为机器编号)

后台发送

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxx",
  "Cmd": "0B",
  "Ack": "00",
  "SN": "xxxxxxxxxxxxxxxxxxxx",
  "Time": {
    "year": xx,      //年
    "month": xx,     //月
    "day": xx,       //日
    "hour": xx,      //时
    "minute": xx,    //分
    "second": xx,    //秒
  }
}
```

设备响应(0BH)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxx",
  "Cmd": "0B",
  "Ack": "00",
  "SN": "xxxxxxxxxxxxxxxxxxxx",
  "Time": {
    "year": xx,      //年
    "month": xx,     //月
    "day": xx,       //日
    "hour": xx,      //时
    "minute": xx,    //分
    "second": xx,    //秒
  }
}
```

设备查询后台方向(1BH) !!未启用

设备发送(1BH)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{
  "ID": "xxxxxxxxxxxxxxxxxxxx",
  "Cmd": "1B",
  "Ack": "00",
  "SN": "xxxxxxxxxxxxxxxxxxxx",
  "Time": {
```

```
        "year":xx,           //年
        "month":xx,         //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}
```

后台响应(1BH)

后台通过jumi/146xxx主题发布(该主题数字部分为机器编号)

```
{
    "ID":"xxxxxxxxxxxxxxxxxxxx",
    "Cmd":"1B",
    "Ack":"00",
    "SN":"xxxxxxxxxxxxxxxxxxxx",
    "Time":{
        "year":xx,           //年
        "month":xx,         //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}
```

远程重启(0CH)

后台发送(0CH)

后台通过jumi/146xxx主题发布(该主题数字部分为机器编号)

```
{
    "ID":"xxxxxxxxxxxxxxxxxxxx",
    "Cmd":"0C",
    "Ack":"00",
    "SN":"xxxxxxxxxxxxxxxxxxxx",
    "data":{
        "DownTime":xx,       //设定掉电时长，单位秒
    }
    "Time":{
        "year":xx,           //年
        "month":xx,         //月
        "day":xx,           //日
        "hour":xx,          //时
        "minute":xx,        //分
        "second":xx,        //秒
    }
}
```

```
}  
}
```

设备响应(0CH)

设备通过jumi/devicecomm0000000000000001主题发布(该主题通过20H指令获得)

```
{  
  "ID": "xxxxxxxxxxxxxxxxxx",  
  "Cmd": "0C",  
  "Ack": "00",  
  "SN": "xxxxxxxxxxxxxxxxxx",  
  "Time": {  
    "year": xx,           //年  
    "month": xx,         //月  
    "day": xx,           //日  
    "hour": xx,          //时  
    "minute": xx,        //分  
    "second": xx,        //秒  
  }  
}
```

远程配置4G模块

注：发送本指令配置完成后设备自动重启。目前仅支持有人7S4、亿佰特4G02、合宙AIR720。

后台发送

```
{  
  "ID": "xxxxxxxxxxxxxxxxxx",  
  "Cmd": "0D",  
  "Ack": "00",  
  "SN": "xxxxxxxxxxxxxxxxxx",  
  "data": {  
    "IPAddress": "xxxxxxx", //远端服务器 IP地址 或 域名  
    "port": xxxx,          //远端服务器端口  
    "mode": "xxx",         //连接类型 "TCP" 或 "UDP"  
  }  
  "Time": {  
    "year": xx,           //年  
    "month": xx,         //月  
    "day": xx,           //日  
    "hour": xx,          //时  
    "minute": xx,        //分  
    "second": xx,        //秒  
  }  
}
```

设备响应

```
{
  "ID": "xxxxxxxxxxxxxxxx",
  "Cmd": "0D",
  "Ack": "00",
  "SN": "xxxxxxxxxxxxxxxx",
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```

设备返回结果

```
{
  "ID": "xxxxxxxxxxxxxxxx",
  "Cmd": "0D",
  "Ack": "01",
  "SN": "xxxxxxxxxxxxxxxx",
  "data": { "Status": xx }, //00 配置失败 01 配置成功
  "Time": {
    "year": xx,          //年
    "month": xx,         //月
    "day": xx,           //日
    "hour": xx,          //时
    "minute": xx,        //分
    "second": xx,        //秒
  }
}
```